Prion diseases comprise several conditions. A prion is a type of protein that can trigger normal proteins in the brain to fold abnormally. Prion diseases can affect both humans and animals. They are sometimes spread to humans by infected meat products. In many cases, the source of the abnormal protein is unknown. The most common form of prion disease that affects humans is Creutzfeldt-Jakob disease (CJD).

Prion diseases are rare. About 300 cases are reported each year in the U.S.
Types of prion diseases include:

- CJD. A person can inherit this condition. In that case, it's called familial CJD. Sporadic CJD, on the other hand, develops suddenly without any known risk factors. Most cases of CJD are sporadic. They tend to strike people around age 60. Acquired CJD is caused by exposure to infected tissue during a medical procedure, such as a cornea transplant. Symptoms of CJD (see below) quickly lead to severe disability and death. In most cases, death occurs within a year.
- Variant CJD. This is an infectious type of the disease. It is related to “mad cow disease.” Eating diseased meat may cause the disease in humans. The meat may cause normal human prion protein to develop abnormally. This type of the disease often affects younger people and is becoming less common.
- Variably protease-sensitive prionopathy (VPSPr). This is also extremely rare. It is like CJD. But the protein is less sensitive to digestion. It is more likely to strike people around age 70 who have a family history of dementia.
- Gerstmann-Sträussler-Scheinker disease (GSS). This type is extremely rare. It occurs at an earlier age, typically around age 40.
- Kuru. This disease is seen in New Guinea. It's caused by eating human brain tissue contaminated with infectious prions. Because of increased awareness about the disease and how it is transmitted, kuru is now rare.
- Fatal insomnia (FI). This is a rare hereditary disorder. It causes trouble sleeping. There is also a sporadic form of the disease that is not inherited.

What causes prion disease?
Normal prion protein is found on the surface of many cells. Prion diseases occur when this protein becomes abnormal and clumps in the brain. It then causes brain damage. This abnormal buildup of protein in the brain can lead to memory problems, personality changes, and trouble with movement. Experts still don't know a lot about prion diseases. But unfortunately, these disorders are often fatal.

How are prion diseases diagnosed?
Prion diseases are confirmed by taking a sample of brain tissue during a biopsy or after death. But given the risks of a brain biopsy, a number of other tests are often done instead. These can help diagnose prion diseases such as CJD or rule out other diseases with similar symptoms. Prion diseases should be considered in all people who have dementia that is quickly getting worse.

How are prion diseases treated?
Prion diseases can't be cured. But certain medicines may help slow their progress. Medical care focuses on keeping people with these diseases as safe and comfortable as possible, despite worsening and debilitating symptoms.

Can prion diseases be prevented?
Properly cleaning and sterilizing medical equipment may prevent the spread of the disease. If you have or may have CJD, don't donate organs or tissue, including corneal tissue. Regulations that govern the handling and feeding of livestock have helped prevent the spread of prion diseases. People should also be careful about hunting and handling wild animals. A small number of US cases have been associated with road and game kill.

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Living with prion diseases
As prion diseases progress, people with these diseases generally need help taking care of themselves. In some cases, they may be able to stay in their homes. But they eventually may need to move to a care facility.

Who is at risk for prion diseases?
You may be at risk for prion diseases if you:

- Have a family history of prion disease
- Eat meat infected by “mad cow disease”
- Are infected by contaminated corneas or medical equipment

What are the symptoms of prion diseases?
Symptoms of prion diseases include:

- Rapidly developing dementia
- Difficulty walking and changes in gait
- Jerking movements of the muscles
- Hallucinations
- Muscle stiffness
- Confusion
- Fatigue
- Difficulty speaking

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The tests include:

- Nervous system and visual exams to check for nerve damage and vision loss
- MRI scans of the brain
- Spinal tap (lumbar puncture) to take samples of fluid from the spinal cord
- Electroencephalogram, a painless test that looks at brain waves by placing electrodes on the scalp
- Blood tests
- Genetic tests to look for known hereditary causes of prion disease. This is especially important if other family members are affected.

Key points about prion diseases

- Prion diseases are very rare.
- Diagnosis is difficult but should be considered in all people whose dementia is quickly getting worse.
- Symptoms can progress rapidly, requiring help with daily needs.
- Prion diseases are always fatal.

Mad cow disease, or bovine spongiform encephalopathy (BSE), is a disease that was first found in cattle. It's related to a disease in humans called variant Creutzfeldt-Jakob disease (vCJD). Both disorders are universally fatal brain diseases caused by a prion. A prion is a protein particle that lacks DNA (nucleic acid). It's believed to be the cause of various infectious diseases of the nervous system. Eating infected cattle products, including beef, can cause a human to develop vCJD.

What is mad cow disease?

Mad cow disease is a progressive, fatal neurological disorder of cattle resulting from infection by a prion. It appears to be caused by contaminated cattle feed that contains the prion agent. Most mad cow disease has happened in cattle in the United Kingdom (U.K.), and a few cases were found in cattle in the U.S. between 2003 and 2006. Feed regulations were then tightened.

In addition to the cases of mad cow reported in the U.K. (78% of all cases were reported there) and the U.S., cases have also been reported in other countries, including France, Spain, the Netherlands, Portugal, Ireland, Italy, Japan, Saudi Arabia, and Canada. Public health control measures have been implemented in many countries to prevent potentially infected tissues from entering the human food chain. These preventative measures appear to have been effective.

What is variant Creutzfeldt-Jakob Disease (vCJD)?

Creutzfeldt-Jakob Disease (CJD) is a rare, fatal brain disorder. It causes a rapid, progressive dementia (deterioration of mental functions), as well as associated neuromuscular disturbances. The disease, which in some ways resembles mad cow disease, traditionally has affected men and women between the ages of 50 and 75. The variant form, vCJD, however, affects younger people (the average age of onset is 28) and has observed features that are not typical as compared with CJD. About 230 people with vCJD have been identified since 1996. Most are from the U.K. and other countries in Europe. It's rare in the U.S.

What is the current risk of getting vCJD from eating beef and beef products from cattle in Europe?

Currently, this risk appears to be very small, perhaps fewer than 1 case per 10 billion servings, if the risk exists at all. Travelers to Europe who are concerned about reducing any risk of exposure can avoid beef and beef products altogether, or can select beef or beef products such as solid pieces of muscle meat as opposed to ground beef and sausages. Solid pieces of beef are less likely to be contaminated with tissues that may hide the mad cow agent. Milk and milk products are not believed to transmit the mad cow agent. You can't get vCJD or CJD by direct contact with a person who has the disease. A few cases were acquired during the transfusion of blood from an infected donor. Most human Creutzfeldt-Jakob disease is not vCJD and is not related to beef consumption but is also likely due to prion proteins.

- Dementia is a general term impaired thinking, remembering or reasoning that can affect a person’s ability to function safely. The term has been replaced by “major neurocognitive disorder” and “mild cognitive disorder” in medical terminology.
- Dementia symptoms may be confused with symptoms of depression or other mood disorders
- Dementia is not a normal consequence of aging.
- The condition has many causes, so patients with signs or symptoms of dementia should get a comprehensive examination to determine the underlying diagnosis.
- There are not direct therapies for most types of dementia, but some conditions, such as normal pressure hydrocephalus, can be treatable. For the remainder, drugs may make symptoms more manageable.

- Symptoms of Dementia
- Dementia may involve a range of progressive symptoms, depending on its cause:
- Mood and personality changes
- Difficulty with words and language
- Poor judgment
- Confusion regarding familiar places and time, date and season
- Inability to concentrate or think clearly
- Difficulties with gait or balance
- Increased daytime sleepiness
- Apathy
- Visual hallucinations

- What are the different types of dementia?
- Dementia may be caused by several other conditions and diseases, including:

- Alzheimer’s Disease
- Alzheimer’s disease is a type of dementia that causes memory loss, confusion, and changes in personality. The course is gradual and includes seven stages. At present, Alzheimer’s dementia is not curable, but medications may help manage symptoms.

- Vascular Dementia
- Vascular dementia (VaD) is the second most common form of dementia after Alzheimer's disease and is caused by damage to brain tissue due to decreased blood flow. Strokes, mini-strokes (TIAs) and congestive heart failure can be the root causes of these conditions.

- Corticobasal Degeneration
- This is a rare atypical Parkinsonian disorder that typically affects one side of the body more than the other. Affected patients may have difficulty seeing and navigating through space.

- Creutzfeldt-Jakob Disease
- A rare, fatal brain disorder, Creutzfeldt-Jakob disease causes a rapid, progressive dementia (deterioration of mental functions) and associated neuromuscular disturbances.

- Dementia with Lewy Bodies
- Dementia with Lewy Bodies is a rare atypical Parkinsonian disorder characterized by an abnormal accumulation of alpha-synuclein protein in brain cells.

Frontotemporal Dementia
Frontotemporal dementia (FTD) is a group of disorders that occur when the nerve cells in the frontal and temporal lobes of the brain are damaged, causing the lobes to shrink.

HIV Dementia
HIV-associated dementia is a serious consequence of HIV infection and is typically seen in advanced stages of the disease.

Huntington's Disease
Huntington's disease is a brain disorder in which brain cells, or neurons, in certain areas of the brain start to break down, eventually leading to emotional disturbances, loss of intellectual abilities and uncontrolled movements.

Other Conditions
Other conditions can cause dementia or dementia-like symptoms, including:

- Stroke or mini-stroke (TIA)
- Congestive heart failure
- Reactions to medications
- Metabolic problems
- Endocrine abnormalities
- Nutritional deficiencies
- Infections
- Brain tumors
- Oxygen deprivation
- Heart and lung problems